

Riders on the Storm^{*}

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Abstract

Digital platforms often subcontract self-employed workers. We develop an equilibrium labor-market model to quantify Spain's 2021 Riders' Law, which sanctions platforms in the online food delivery sector for hiring self-employed couriers, known as riders. Riders in the model trade off flexibility and employability as subcontracted against higher wages and guaranteed hours of work as employees. We find that the reform increases the employee share by 2 percentage points but that this rise falls short of absorbing job losses among the self-employed. Overall, riders and consumers experience welfare losses due to lower average hourly wages and employment in the sector as a whole as a result of the sanctions. However, we find that redistributing sanction revenue as lump-sum transfers to riders offset those losses.

Keywords: Riders, Food delivery platforms, Self-employed, Employees

JEL: J21, J60

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1 Introduction

Like a dog without a bone, an actor out on loan, riders on the storm (**Riders on the Storm, The Doors**)

The digital platform economy is expanding rapidly, with estimates of the share of online gig workers in the global labor force ranging from 4.4 to 12.5 percent (Datta et al., 2023). Because the degree of subordination of workers to platforms is difficult to determine, platforms often classify these workers as independent contractors instead of as employees. Some pundits defend this practice on the grounds that self-employment not only offers workers greater flexibility in terms of working hours but also facilitates their employability. Yet, critics observe that the status of independent contractors takes away from workers' essential social protections, such as compensation for the number of hours actually worked and collectively bargained wages. Legislators worldwide have responded to these critics by proliferating new regulations of platform workers' working conditions.¹

The Spanish 2021 Riders' Law (RL, hereafter), provides a unique policy experiment to study the macroeconomic effects of regulating labor contracts in the online food delivery sector, which currently is the second-largest platform industry globally after taxi services. The RL requires these platforms to treat their couriers, commonly known as *riders*, as employees instead of independent contractors. However, two major platforms resisted the new regulations by successive appeals to courts and, in the meantime, continued using subcontractors after the reform. Importantly, in doing so, platforms in Spain remained loyal to the supranational business models implemented by their parent companies which, therefore, provides us with plausible exogenous variations in compliers and defiers of the RL. As a result, we observe the two types of business models operating in parallel both before and after the reform, with the RL increasing marginal costs for defier platforms via the imposition of administrative fines. Thus, as in the lyrics of the classic song by The Doors heading this section, our riders, like theirs, seem to be weathering a storm full of uncertainties, this time related to their labor conditions.

To evaluate the consequences of the RL, we build an equilibrium model of the labor market that accounts for the above-mentioned key policy trade-offs. Platforms with different business models provide food orders to customers in an imperfectly competitive goods market. The different business models give rise to two types of rider jobs. First, regular (R) platform jobs provide riders with employee status. These jobs require employers to pay social security contributions, offer compensation for fixed working hours,

¹For example, Australia (Fair Work Legislation Amendment Act 2024), Belgium (Wet van 28 november 2022), California (Assembly Bill 5 in 2019), Chile (Ley N° 21,431 in 2022), EU parliament (Directive 2024/2831), France (Loi 2016-1088), Italy (Legge 128/2019), Massachusetts (Ballot Question 3 in 2024), Mexico (Ley Federal del Trabajo 2024), Portugal (Lei 13/2023), South Korea (Employment Insurance Act 17,859 in 2021), or Uruguay (Ley N° 20,396 in 2025).

and set hourly wages through Nash bargaining between the employer and a trade union. Given that the hiring and vacancy-opening processes of these platforms are similar to those in the non-platform economy (entailing hiring and firing costs, etc.), search frictions impede workers from being immediately hired by R platforms. Second, there are instantaneously available casual (C) platform jobs, where riders sign on to the platform as self-employed. Riders take care of their payroll taxes, freely choose their hours worked, and are paid per delivered order instead of by hour worked. The platform aligns the supply of hours with any demand for orders by means of the utilization rate of riders, i.e., the amount of working time they spend delivering rather than waiting for orders. Riders with idiosyncratic preferences regarding hours worked choose where to search for jobs and how many hours to work when self-employed. They always have the option of joining a C platform and then search on-the-job for R platform jobs if these are their preferred option. Finally, riders also have an outside option capturing riders' available job opportunities in the non-platform labor market.

We estimate the model parameters to fit post-reform moments regarding hours worked, wages, market shares, and firms' provisions for RL-related legal risks. To do so, we use firms' annual reports, aggregate statistics from the Spanish food delivery sector, and international estimates of the labor supply elasticity to the platform sector. As data on wage premia between platforms and hours worked by platform are not available for Spain, we combine this data with microdata from a representative convenience survey. The survey uncovers three regularities relevant to the policy trade-offs discussed above: (i) riders with similar characteristics earn a log hourly wage premium of 0.17 in R jobs relative to C jobs, (ii) a substantial fraction of riders in both platforms overlap in their hours worked, and (iii) some C riders work a relatively high number of hours. The model rationalizes these facts by riders who wish to work long hours sorting into C jobs to achieve higher earnings by working many hours, and other riders searching for R jobs due to their wage premium. Yet, due to search frictions, those who fail to find an R job instantaneously choose instead to join a C platform in the meantime. In fact, due to a low utilization rate of riders' hours on the C platform, around three-fourths of C riders earn less than those in the R platform despite working longer hours. Hence, *ceteris paribus*, a reform that expands employment at the R platform offers potential benefits to riders.

The goal of the RL is to expand R jobs through fines on defier C platforms. We compute the size of the sanctions from their provisions for RL-related legal risks between 2021 and 2024, which amount to 3.2 percent of their net revenue. Importantly, instead of being fixed costs, these sanctions are proportional to the number of bogus self-employed riders due to size-dependent penalties and back-payment of avoided Social Security taxes. Therefore, sanctions increase the marginal cost of producing orders by C platforms, raising their relative prices and shifting consumers' order demand to R platforms.

Counterfactual simulations show that the reform accounts for half of the increase in the market shares of compliant platforms observed in the data after the RL (see Figure 1 below). To satisfy additional demand, the employment share in the R platform expands by 2.0 percentage points, absorbing 50 percent of the C job losses resulting from the reform. As mentioned above, in partial equilibrium, the reallocation towards the R platform would be beneficial for most C riders. However, once one accounts for general equilibrium effects, this is no longer the case. The intuition behind the result is that C riders' hours utilization rate declines yet further since those riders switching to the R platform produce more orders than at their previous C jobs, leaving stayers with less demand and, hence, a lower utilization rate. This effect implies that their hourly pay to fall by 5 log points. Importantly, we show that the model is consistent with evidence from the U.S. platform sector on how utilization rates respond to demand-driven pay changes and on labor supply elasticities (Hall et al., 2023; Caldwell and Oehlsen, 2026). As conditions at the C platform get worse, employee riders' outside option in bargaining with the R platform also worsens, resulting in a reduction of 3 log points decline in their hourly wages. All in all, we find that the food delivery sector shrinks by about 2 percent while its average net hourly wage drops by 4.5 log points, leading to a reduction of both riders' and consumers' welfare.²

Finally, we study the effects of complementing the sanctions with transfers to riders. In the first experiment, the revenues generated by fines are returned to the riders through lump-sum rebates. This policy reverses the previous welfare loss and achieves a welfare gain for the average rider of 0.47 in terms of consumption-equivalent variation. However, some riders still experience welfare losses. Moreover, consumer welfare falls because lump-sum transfers do not avoid the shrinkage of the food delivery sector. In addition, we simulate the effect of an alternative tax policy reducing employers' social security contributions in the R platform, which the RL aims to expand. Our finding is that tax cut ameliorates the welfare losses but does not eliminate them.

1.1 Related literature and outline

Our paper contributes to a growing empirical literature on the characteristics of the online gig economy which addresses the difficulty of measuring platform work arrangements through either conventional administrative data, specific surveys, or field experiments (Mas and Pallais, 2017; Katz and Krueger, 2019; Boeri et al., 2020; Bracha and Burke, 2021; Garin et al., 2022, 2023). In this respect, we combine an own-elaborated convenience survey with representative aggregate information from the food delivery sector.

²Given that employment has increased by 40 percent since the pandemic (see Sections 2 and 6 below), these negative effects of the RL should be interpreted as deviations from a growing trend in household demand for online food delivery, which our counterfactual steady-state comparisons hold constant.

Similar to us, there is a recent literature that highlights the value that digital platforms provide to workers through flexible working arrangements. [Chen et al. \(2019\)](#) study the case of Uber drivers and show that they value flexibility in their working hours, with their surplus being higher than in alternative work arrangements which offer less flexibility. [Stanton and Thomas \(2025\)](#) analyze a platform environment where buyers post one-time projects and workers compete for these projects by posting wages. They find that imposing traditional employment regulations on short-term platform work reduces consumers' and workers' welfare. Key to this finding is that the regulation decreases demand for their services. We add to this literature by studying a setup where flexible platform jobs coexist with less flexible ones, highlighting the trade-offs faced by agents in this market. In particular, we show that regulating firms with independent contractors may have negative spillover effects on employees' wages when these are bargained.

A key adjustment mechanism in our model is the utilization rate of riders in C jobs. Endogenous utilization rates also play a key role in the house-broker market ([Hsieh and Moretti, 2003](#)) and the Uber driver market ([Hall et al., 2023](#)), with product and labor demand shocks lowering these rates due to labor-supply responses. Related to this literature, we show that a shock shifting orders from the C platform to the R platform decreases the utilization rate of those riders who remain operating in the C platform after the reform insofar as total consumer demand is sufficiently inelastic.

Our paper also relates to the literature that uses structural equilibrium models to study the informal economy, which resembles independent contractor arrangements in the platform economy. Part of this literature builds two-sector models that feature a frictional and a frictionless sector, such as [Zenou \(2008\)](#) and [Satchi and Temple \(2009\)](#), who model an economy with homogeneous workers who always prefer formal jobs. Instead, our approach is similar to [Albrecht et al. \(2009\)](#), [Haanwinckel and Soares \(2021\)](#), and [Feng et al. \(2024\)](#) by allowing for ex-ante worker heterogeneity to generate the possibility that some workers may prefer C jobs. Different from these papers, our sorting mechanism is based on the taste for hours worked rather than by ability. In this fashion, self-selection of workers into sectors has key policy implications arising from the interaction of search frictions in the R sector and utilization rates in the C sector. Moreover, while these studies focus on the effects of taxes/transfers on formal-sector jobs, we study the effects of regulating C jobs and show that this policy leads to spillover effects on wages and employment in R jobs through the role of outside options in collective bargaining.

The rest of the paper is organized as follows. Section 2 provides institutional background about food delivery platforms in Spain, plus details of the 2021 Riders' Law. Section 3 describes the data sources and draws stylized facts about the riders' labor market outcomes. Section 4 lays out the quantitative model. Section 5.1 describes its

calibration. Section 6 shows the counterfactual results from several policy experiments. Section 7 concludes. An Appendix gathers further evidence discussed in the main text.

2 Institutional Background

As in many other developed economies, the Spanish online food delivery sector has expanded rapidly over the past decade. This expansion has taken place in a context of uncertainty about the correct employment status of riders due to the legal difficulty of determining the degree of workers’ subordination to platforms. Consequently, working as an independent contractor instead of as a dependent employee was the most prevalent form of employment in the food delivery sector prior to the RL.

Court records illustrate what onboarding as a self-employed rider looks like (TSJ Andalucía, 2025). Applicants must be of legal age, own a vehicle, and register as self-employed with the tax and social security authorities. Riders then sign a commercial services contract with the platform and receive orders while connected to the app. Such employment relations are governed by Spain’s Self-Employed Workers’ Statute (Law 20/2007, or *Estatuto del Trabajo Autónomo*), which offers several advantages to platforms in comparison to hiring employees that are regulated by the Labor Code (Royal Legislative Decree 2/2015, or *Estatuto de los Trabajadores*): Platforms do not pay payroll taxes, are not subject to labor law on overtime schedules and dismissal costs, and avoid unionization of their workers. Moreover, unlike employees, self-employed riders have no guaranteed working hours and are paid for completed orders rather than at a fixed hourly wage. As a result, platforms shift the entire risk of insufficient demand to riders, who commonly face working days with long unpaid breaks (ILO, 2024).

These practices fueled an intense debate about the appropriate regulation of work arrangements that culminated in the Spanish parliament enacting the RL (Royal Legislative Decree 9/2021) in September 2021. Regarding the motivations of policy-makers, the preamble of the RL states that the advantages of information and communication technologies “are perfectly compatible with the objective of labor law to re-balance the interests of the different agents in this sector”, and that these regulations should “guarantee that technological change provides positive effects equitably”. To achieve these goals, the RL introduces: (i) the requirement that platforms disclose the rules underlying algorithmic decisions that affect workers’ labor market conditions, and (ii) the presumption that riders are employees rather than independent contractors.

Our analysis hinges on two key institutional features. First, by only introducing the presumption of an employee status for riders, there remains some legal uncertainty

regarding their correct classification. Consequently, some platforms defied the new regulation by claiming in different courts that they did not exert powers of organization over their riders. Second, food delivery platforms are owned by multinational companies that decide their business models on a supranational scale; hence, their practices for labor contracts are plausibly orthogonal to the RL. As a result, the sector features riders with different employment statuses both before and after the reform.

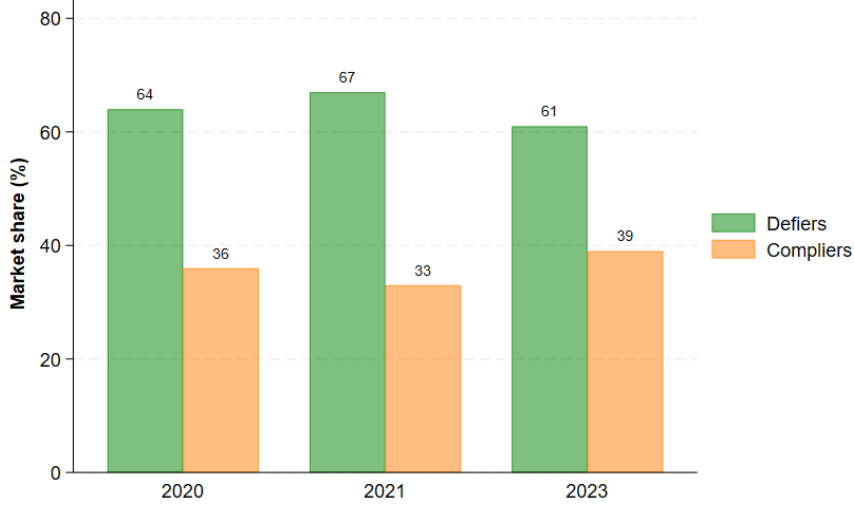
In particular, Just Eat, one of the largest platforms that holds one-third of the market share, had already largely shifted to hiring riders as employees in its European markets before the RL, including Spain ([Just Eat Takeaway.com, 2021](#)); hence, we label it as “complier”. The collective agreement between Just Eat and the main trade unions in the sector describes these employee work arrangements ([Just Eat–CCOO–UGT, 2021](#)).³ Riders sign an employment contract that sets guaranteed ordinary hours, caps additional hours, and establishes wages based on hours worked. The remaining three dominant players (Glovo, Uber Eats, and Deliveroo), with a joint market share of two-thirds, had hired most of their riders as independent contractors globally and continued doing so in Spain after the RL, so we coin them “defiers”.⁴ Importantly, the legal arguments of the defiers did not convince the government, which responded with administrative sanctions for which such platforms have provisioned approximately €923.5 million for the resulting legal risks. We interpret these provisions as the present discounted expected value of sanctions, which represents about 3.2 percent of defiers’ net revenue (see Appendix A).

Indeed, market share data after the enactment of the reform is consistent with a deteriorating relative advantage of the defier platforms. Figure 1 shows that their market share declined by 6 percentage points within two years of the reform, despite having expanded previously. Partly, the decline results from the exit of Deliveroo and other small platforms from the Spanish market. Though shifting employment to platforms hiring employees is most likely consistent with the goal of the law’s drafters, some organized labor groups have opposed the reform. For example, according to one of the main workers’ associations (*Asociación Autónoma de Riders, AAR*), about 65 percent of its members dislike the new regulations, mainly due to the rigidity of hours of work in the *R* platforms.

³For details, [Leschke and Scheele \(2024\)](#) provides a summary of the work arrangements in the food delivery sector by contract status across four European countries, including Spain.

⁴For example, Delivery Hero SE, the parent company of Glovo, reported in their global annual report that “the majority of the riders are engaged as freelancers” and their “Public Affairs team globally are in constant exchange with national and local authorities to promote a regulatory framework (self-employment) that works for all involved parties” ([Delivery Hero SE, 2022](#)).

Figure 1: Market shares in the Spanish food delivery sector



Source: We use information from Measurable AI (Spain Food Delivery Market Overview) for 2020 and 2021. For 2023, we use data from Dashmote (Spain's Food Delivery Market: An Analytical Overview)
 Note: The Figure displays the revenue market share in the Spanish food delivery sector.

3 Data and Stylized Facts

Spain has no administrative data that allows to distinguish wages and hours of riders working for different platforms. Thus, we conduct our own online convenience survey. Reassuringly, a comparison of this dataset with aggregate data from alternative sources shows that our survey respondents closely resemble the broader rider population in terms of key sociodemographic attributes, supporting that selection bias in our sample is minor. We document three main findings from the survey: (i) higher net hourly wages are earned by R riders, controlling for sociodemographics, (ii) relatively long hours worked are only a feature of C riders, and (iii) short hours are common for both types of riders.

Data. Through personal contacts with staff at both complier and defier platforms, we distributed an anonymous convenience survey to their riders via Google Forms in September and October 2023. The survey circulated as a shareable link among riders at the three main platforms (Just Eat among compliers, and Glovo and Uber Eats among defiers). Consistent with the market shares reported in Table 1, two-thirds of the contacted riders worked for defier platforms and one-third for complier ones.

The questionnaire was organized around three blocks: (i) general information about the worker (age, gender, educational attainment, nationality, availability of a work permit, number of platforms, and tenure); (ii) information about their net hourly wages and daily hours worked; and (iii) the platform they work for. As regards wages and working hours, the riders were asked to report their monthly earnings and daily hours of work around

Table 1: Descriptive statistics

	Convenience survey (2023)	F. La Caixa (2021)	Adigital (2020)
<i>Workers' characteristics</i>	Mean	Mean	Mean
Age	27.3	30.5	29.3
Gender (Male)	0.86	0.86	0.89
Education (Upper)	0.46	0.41	0.39
Nationality (Foreign)	0.77	0.75	0.72
Work Permit (Yes)	0.82	0.65	0.75
No. of platforms (2023)	1.3		
Tenure (years)	1.5		
<i>Wages and hours</i>			
Net hourly wage (Euros)	5.6		
Daily hours	4.6		
<i>Platform employment shares</i>			
Defiers	0.76		
Compliers	0.24		

Note: The convenience survey data collected information on 162 riders via Google.form corresponding to the period between September and October 2023. The Fundación La Caixa survey (Longoni et al., 2023, 2025) provides socio-demographics from 392 riders between April - May 2021. The Adigital (2020) survey contains information about 1,852 riders during the period January-February 2020. Net hourly wages are computed by subtracting supply/ maintenance costs and social security contributions borne by self-employed riders from gross hourly wages

the dates of the survey, on top of their estimated monthly supply and maintenance costs (bike, cell phone, backpack, etc.) and social security contributions in case being self-employed, which altogether represent 32 percent of gross wages on average (see Appendix B for details).⁵ With that information at hand, we computed hourly net wages and daily working hours. In total, 320 riders received the questionnaire, out of which 162 responded all the questions. Since approximately 30,000 riders were active in 2023 according to the AAR, our sample represents about 0.5 percent of the rider population at that time. To assess the representativeness of the survey, we compare the sociodemographic composition to a larger survey conducted by the Spanish Association of Digital Economics (Adigital, 2020), which collected information for a representative sample of 1,852 riders in early 2020. Similarly, we compare the composition to a survey of almost 400 riders carried out by Fundación La Caixa during the spring 2021 (see Longoni et al., 2023, 2025) Unlike ours, both of these larger surveys lack information on wages, hours worked, and the specific platforms riders work for.

Table 1 summarizes the main descriptive statistics of the Adigital (2020) survey, The Fundación La Caixa survey, and our convenience survey. Respondents in the convenience survey closely resemble the broader population of riders captured in the larger surveys

⁵The wording of the question about wages explicitly excluded income taxes.

across key sociodemographic dimensions, including age, gender, education, nationality, and work permits.⁶ Riders are predominantly young, male, foreign, and their tenure is 18 months. Moreover, the average rider typically works for a single platform. Interestingly, between 18 and 35 percent of riders are migrants without a work permit who work for C platforms through subcontracted self-employment accounts. Employing them is unlawful under either arrangement, but the subcontracting channel makes detection and enforcement far less likely than hiring them directly as employees would. This may partly explain the resistance of the C platforms to comply with the RL.

Stylized facts about hours and wages. As described above, an advantage of being hired as an employee is being paid by the hour with a wage that is set by collective bargaining. Moreover, R platforms pay payroll taxes. To understand the effect of the employee status on net hourly wages, we run the following OLS regression:

$$\ln w_i = \beta_0 + \beta \mathbb{I}_{C=1} + \gamma \mathbf{X}_i + \varepsilon_i$$

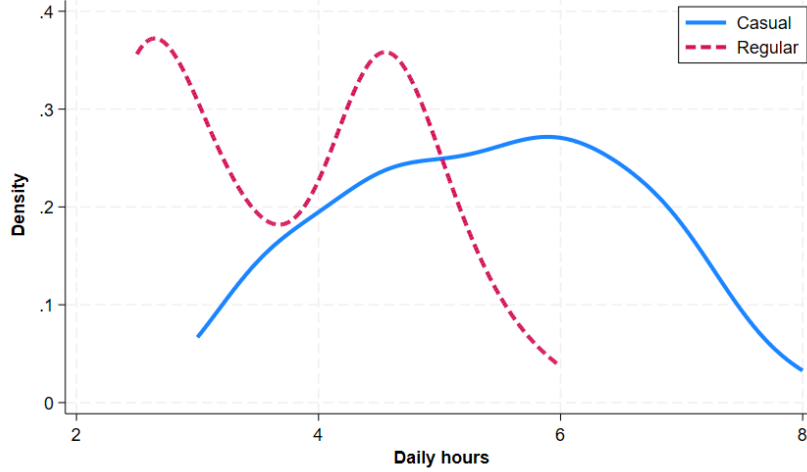
where $\ln w_i$ is (logged) net hourly wage, $\mathbb{I}_{R=1}$ is a dummy variable capturing if a rider works in the C platform (Glovo or Uber Eats), and $\mathbf{X}_i$ is a vector of socio-demographic covariates that includes age, gender, nationality, work permit, tenure, and education. We find that, on average, the hourly wage among C riders is about 17.2 log points lower than that of R riders with similar observable characteristics ($\beta = -0.172^{***}(0.033)$). This is consistent with trade unions being effective in extracting rents.

Note that a sizable share of riders (76 percent) opt for C jobs despite their lower wages. To understand this fact, Figure 2 displays the densities of daily hours worked in both types of jobs, where four findings stand out. First, there is a large dispersion in daily hours across riders, suggesting large heterogeneity in their time availability, which is consistent with reports from the Adigital (2020) survey. Second, the distribution of hours worked in C jobs exhibits higher dispersion than in R jobs; in particular, the density of hours worked of the former jobs is smooth, while it is bimodal for the latter.⁷ Third, C jobs mainly provide upward flexibility in hours worked: while R riders work less than 6 hours per day, a significant share of C riders exceed this threshold, so that their average daily hours worked C is higher than in the R sector (5.4 h. vs 3.7 h.), a result that also holds after controlling for observables. Fourth, the distribution of hours worked has a significant mass at low hours worked in both platforms, typically below 4 hours. We

⁶Unfortunately, since the microdata of the Adigital and Fundación La Caixa surveys are not available, we cannot provide standard statistical tests about differences in means.

⁷Just Eat offers two main [job contracts](#): one of 12 weekly hours during weekends and another of around 24 hours during the entire week. For tractability, we later assume in the model that R platforms offer a single work schedule that represents a weighted average of the two contracts.

Figure 2: Distribution of hours worked



Note: The Figure displays the density of hours worked for workers with a casual and regular platform job using a Gaussian Kernel estimate.

interpret this fact as being consistent with search frictions in the R platform that ration the access of riders with strong preferences for these jobs due to their wage premium.

One final comment is worth noting before turning to the model. A recurring finding in the literature is that self-employed gig workers usually work fewer hours than those established by an ordinary full-time contract (see, e.g., Boeri et al., 2020). This is also the case in our survey: the average self-employed rider only works 5.4 hours per day. Other time commitments such as education, taking care of a family member, or a second job may explain the low average. What our findings show is that, *within the same occupation*, the distribution of hours worked among the self-employed differs from that of employees in the ways described in the previous paragraph. These findings are consistent with data from Chile, a unique country in providing representative data for the transportation platform sector (Bamieh et al., 2026), as well as with evidence from other sectors.⁸

4 Quantitative Model

Environment. We evaluate the effects of the 2021 RL through the lens of a quantitative discrete-time steady-state equilibrium model that incorporates the main trade-offs described above between the subcontractor and employee status in the platform sector. The options for potential riders are: (i) to start working as self-employed at a casual (C)

⁸Using data from the Spanish Labor Force survey, we restrict the sample to self-employed and private employees with open-ended contracts. Then, we regress log ordinary hours on self-employment status, controlling for 1-digit occupation, 1-digit sector, and region fixed effects. We find that self-employed workers work 0.25 log points more hours than employees. We also find that residual hours are more smoothly distributed among the self-employed than among private employees.

jobs platform, where they freely choose their optimal working hours; (ii) to search for a job with a regular (R) jobs platforms, where they obtain an employee contract and get paid a collectively bargained hourly wage for a fixed number of hours; and (iii) an outside option, representing job opportunities amenable to riders in the non-platform economy. We do not model platforms' choice of labor contract, which we attribute to their supranational business practices. Furthermore, riders are ex-ante heterogeneous in their tastes for hours worked, determined by parameter ϵ (to be defined below). Finally, platforms sell orders to a representative consumer who views their respective services as imperfect substitutes. Thus, the model features both types of platforms operating in equilibrium with their market shares being endogenously determined.

Production technology. We consider two representative groups of platforms with identical production technologies, which are characterized by the type of job they offer, $j = C, R$. Each produces orders, o_j , with a linear technology in labor:

$$o_j = \int_0^{n_j} A h_j^p dj, \quad (1)$$

where n_j denotes employment, h_j^p are the productive hours worked (defined below), and A captures productivity per hour.

Labor market for regular jobs. Employee riders earn *per hour* wages set by collective bargaining and work fixed hours $\bar{h}_R$, reflecting the contractual limits on working time.⁹ As riders are regular employees who are paid for fixed hours, platforms, as any other firm in the economy, have to screen job applicants and make sure to hire workers consistent with demand conditions. We model this hiring process of R jobs as standard Diamond-Mortensen-Pissarides search frictions. Searchers and the R platform meet according to a matching technology that determines the contact probabilities of riders and vacant jobs. To ensure that these probabilities are bounded between 0 and 1, we adopt the CRS matching function proposed by [Den Haan et al. \(2000\)](#), namely, $m(s, v) = sv/(s^\iota + v^\iota)^{1/\iota}$, where ι is a matching parameter. Hence, the contact probabilities for job seekers and for vacancies are $p(\theta) = m(s, v)/s$ and $q(\theta) = m(s, v)/v = p(\theta)/\theta$, respectively, where $\theta = v/s$ is labor market tightness, given by the ratio of vacancies, v , to searchers, s , where the latter encompasses both non-riders and on-the-job searchers.

Labor market for casual jobs. The labor market for C jobs is fundamentally different from that of employees. Instead of having contractual hours and being paid by the hour, riders with a C job choose their hours supplied and are paid *per order* produced. To the platform, the cost of an inactive rider is zero under this piece-rate arrangement. Thus, the

⁹Figure 3 below shows that riders who prefer short hours search for R jobs due to their wage premium. Thus, allowing for part-time schedules on the R platform would not alter worker sorting across jobs.

C platform keeps positions open for all riders willing to work, who can then take up these jobs instantly by registering in the app as self-employed. However, C riders may spend only a fraction of their time delivering orders when the supply of hours exceeds demand. We refer to the ratio between productive and actual work times as the utilization rate of the C platform, namely, $h_C^p = \varphi h_C$, where $0 < \varphi \leq 1$ is the endogenous utilization rate strictly lower than one when there is excess supply of hours. Hence, the market-clearing condition for the C platform is

$$o_C = \int_0^{n_C} A h_j^p dj = n_C \varphi A \int h_C(\epsilon) dG^C(\epsilon), \quad (2)$$

where G^C is the cumulative distribution function (CDF) of the riders' taste parameter ϵ for the hours worked in these jobs, so that $\int h_C(\epsilon) dG^C(\epsilon)$ corresponds to the average hours worked per rider.

Demand for food delivery orders. We model the demand for food delivery orders through a representative consumer with exogenous income m who chooses the number of orders o_j from each platform by maximizing the following CES utility function subject to a linear budget constraint:

$$\mathcal{U}_c = \max_{o_C, o_R} \left(\sum_{j \in \{C, R\}} (s_j o_j)^{\frac{\mu-1}{\mu}} \right)^{\frac{\mu}{\mu-1}}, \quad (3)$$

$$\text{s.t. } p_C o_C + p_R o_R = m. \quad (4)$$

where $\mu > 1$ is the elasticity of substitution between orders provided by C and R platforms, and s_j are utility shifters capturing the importance of non-price characteristics of food delivery services in platform j .¹⁰ Note that, for tractability, we abstract from other potential substitute goods in the utility function (e.g., cooking at home). The first-order conditions (FOC) of the consumer problem yield the relative inverse demand of orders:

$$\frac{p_C}{p_R} = \left(\frac{s_C}{s_R} \right)^{\frac{\mu-1}{\mu}} \left(\frac{o_R}{o_C} \right)^{\frac{1}{\mu}}. \quad (5)$$

Since consumption choices only depend on the relative utility shifter, we normalize $s_C = 1$.

Firm's problem. Firms choose the number of orders produced to maximize flow profits, having internalized their downward-sloping demand functions and taking as given

¹⁰Imperfect substitution between goods captures the fact that food delivery platforms are not price takers but operate in an oligopolistic sector. Media evidence also indicates that Spanish consumers are not indifferent between platforms, showing clear preferences between, for example, Glovo and Just Eat based on restaurant availability or app functionality. Glovo focuses on multi-product delivery, whereas Just Eat is valued for its network of local restaurants and its promotions (El Español, 2021).

both the riders' labor supply decisions and their competitors' decisions:

$$\begin{aligned} & \max_{o_j} (p_j(o_j) - mc_j) o_j, \\ \text{s.t. } & p_j(o_j) = \left(\frac{s_j}{s_{-j}} \right)^{\frac{\mu-1}{\mu}} \left(\frac{o_{-j}}{o_j} \right)^{\frac{1}{\mu}} p_{-j}, \end{aligned}$$

with mc_j denoting the marginal cost of each platform j . The FOC yields the standard Lerner condition for the price as a markup over the marginal cost:

$$p_C^* = \frac{\mu}{\mu - 1} \cdot mc_C, \quad \text{and} \quad p_R^* = \frac{\mu}{\mu - 1} \cdot mc_R, \quad (6)$$

where the markup is determined by the elasticity of substitution between orders μ . Hence, (5) and (6) highlight that, besides the demand shifters s_j , relative market sizes are determined by the marginal costs of platforms to which we turn next.

Determinants of marginal costs. The specific structure of the riders' labor market implies that marginal costs are different for platforms according to whether they hire workers as independent contractors or employees. Platforms offering C jobs pay w_C to their riders per order produced, which is chosen to be the numeraire of the economy and normalized to $w_C = 1$. In addition, C platforms may have to pay administrative sanctions. Consistent with the increase of these with the number of misclassified riders due to size-dependent labor-law violation penalties and back-payment of taxes, we interpret these sanctions, Γ , as adding to firms' marginal costs. Thus, given that these C employers do not pay social security contributions, their marginal costs are given by:

$$mc_C = w_C(1 + \Gamma) = (1 + \Gamma). \quad (7)$$

Equation (7) implies that, *ceteris paribus*, imposing administrative sanctions will raise prices of the C platform and, thus, redirect food delivery orders towards the R platform.

Turning to the marginal costs of the R platform, it pays an hourly wage $w_R(\bar{\epsilon}_R)$ plus payroll taxes τ_f . Different from C riders, R riders enjoy bargaining power through a union that bargains with the firm, taking into account the average taste parameter of its members, denoted as $\bar{\epsilon}_R$. To avoid the complication of modeling multi-worker bargaining arising from the incentives of firms to over-hire (Stole and Zwiebel, 1996; Elsy and Michaels, 2013), which is not relevant to the main point of our analysis, we assume that there is an HR division within each R platform that is disconnected from the production side and treats the contribution of each rider to the firm individually.¹¹ In particular,

¹¹An equivalent micro-foundation for abstracting from multi-worker bargaining is to assume that the firm hires labor from intermediate firms, each one with one worker, which play the role of the indirect provider of labor services to the platforms, as in Marimon and Zilibotti (2000).

to cover labor costs, the production side of the platform transfers a fee c^{HR} to the HR division for each hour worked by a rider, which is then used to pay the corresponding gross hourly wage $(1 + \tau_f)w_R(\bar{\epsilon}_R)$. Thus, the value for this HR division of providing an additional rider to the platform becomes:

$$J_R(\bar{\epsilon}_R) = c^{HR}\bar{h}_R - (1 + \tau_f)w_R(\bar{\epsilon}_R)\bar{h}_R + \beta(1 - \delta)J_R(\bar{\epsilon}_R), \quad (8)$$

where β is the time discount factor, and δ is an (exogenous) job destruction rate. Importantly, due to the presence of search and matching frictions, the HR division incurs a flow vacancy posting cost κ when searching for a rider. Thus, the value of a vacancy is given by:

$$J_R^v(\bar{\epsilon}_R) = -\kappa + \beta \left[q(\theta)J_R(\bar{\epsilon}_R) + (1 - q(\theta))J_R^v(\bar{\epsilon}_R) \right].$$

In equilibrium, the R platform posts vacancies until the value of the vacancy becomes zero, implying that:

$$\kappa = \beta q(\theta)J_R(\bar{\epsilon}_R). \quad (9)$$

Plugging (9) into the value of a filled job given in (8) yields:

$$c^{HR} = (1 + \tau_f)w_R(\bar{\epsilon}_R) + \left(1 - \beta(1 - \delta)\right) \frac{\kappa}{\beta q(\theta)\bar{h}_R}. \quad (10)$$

As mentioned above, the labor costs of an R platform consist of paying a fee of $c^{HR}\bar{h}_R n_R$ to its HR division. Moreover, the production technology in (1) yields that the total number of hours worked by R riders is given by $\bar{h}_R n_R = o_R/A$. Consequently, its marginal cost of orders is given by:

$$mc_R = \frac{c^{HR}}{A}, \quad (11)$$

which highlights that, resulting from the search and matching friction, R platforms' marginal costs are driven not only by hourly wages and payroll taxes, but also by the expected discounted cost of posting vacancies.¹²

Finally, wages for R riders solve the standard Nash-bargaining problem:

$$w_R(\bar{\epsilon}_R) = \arg \max \left\{ \left[W_R(\bar{\epsilon}_R) - U(\bar{\epsilon}_R) \right]^\eta \left[J_R(\bar{\epsilon}_R) \right]^{1-\eta} \right\}, \quad (12)$$

where η is the riders' bargaining weight, and $U(\bar{\epsilon}_R)$ is the outside option value of not

¹²Note that, unlike the marginal cost of the C platform, productivity per hour (A) enters the marginal cost of the R platform as it pays riders *per hour* worked, while the C platform pays *per order* produced.

working for the R platform (for a rider with the average taste parameter, $\bar{\epsilon}_R$).

Summing up, differences in marginal costs and, hence, in demand for orders between the two types of platforms arise from: (i) C platforms having to pay fines after challenging the RL, and (ii) R platforms having to pay bargained wages with a union plus social security contributions, as well as internalizing the costly labor market search process. As discussed earlier in Section 3, the resulting bargained wage is higher in the R platform than the labor compensation in the C platforms. Hence, in the absence of administrative fines, C platforms set lower prices, which would allow them to capture a larger share of the online food delivery market.¹³

Riders' preferences There is a unit mass of workers who are ex-ante heterogeneous in their distaste of work, reflecting differences in caregiving responsibilities, educational commitments, and other time constraints. The taste parameter leisure, ϵ , is distributed as a left-truncated normal distribution, $\epsilon \sim N(\mu_\epsilon, \sigma_\epsilon^2, 0, \infty)$, across workers. Individuals are endowed with $\tilde{h}$ units of time per period, and their utility depends on both labor income, y , and the share of time allocated to leisure, $1 - h/\tilde{h}$. Specifically, the flow utility of riders is given by:

$$u(y, h) = \ln(y) + \epsilon \ln(1 - h/\tilde{h}), \quad h \in [0, \tilde{h}].$$

Modeling riders' utility as dependent on labor income captures that they are hand-to-mouth and hence do not save. Recall that riders earn labor income that depends on hours worked in R and on orders produced in C , and that the latter are not remunerated for their wait time, leading to an hourly compensation of $w_C \varphi A$. Moreover, as self-employed, they pay their own payroll taxes τ_C . Lastly, non-rider job seekers receive a flow value b_U from working $\bar{h}_U$ hours in an outside option representing alternative job opportunities. Therefore, consumption/income for each employment status becomes:

$$\begin{aligned} y_U &= b_U \bar{h}_U && \text{if searcher,} \\ y_R &= w_R(\bar{\epsilon}_R) \bar{h}_R && \text{if employed in } R, \\ y_C &= (1 - \tau_c) w_C \varphi A h_C && \text{if employed in } C, \end{aligned}$$

¹³In our setup, note that these lower prices directly affect consumers. In practice, however, platforms may also compete in the fee they charge to restaurants. As a result, C platforms were able to sign more restaurants to their delivery services. We do not model this intermediate step. Yet, assuming that restaurants operate under perfect competition, they will pass any reduced fees to consumers in the form of lower meal prices.

Accordingly, the period-by-period utility functions become:

$$u_C(\epsilon) = \max_{h_C} \ln(y_C) + \epsilon \ln(1 - h_C/\tilde{h}) \quad (13)$$

$$u_R(\epsilon) = \ln(y_R) + \epsilon \ln(1 - \bar{h}_R/\tilde{h}), \quad (14)$$

$$u_U = \ln(b_U) + \epsilon \ln(1 - \bar{h}_U/\tilde{h}). \quad (15)$$

Note that the key determinants of the flow value of the outside option, b_U and $\bar{h}_U$, are exogenous in our model. That is, we consider that the platform sector does not influence equilibrium outcomes in the rest of the economy due to its relatively small size.

Labor supply choices by riders. Let $U(\epsilon)$ denote the value of potential riders currently not employed as a rider (non-rider searchers, hereafter). These potential riders may be either non-employed or they may work outside of the platform sector and wish to become a rider. Hence, their value solves:

$$U(\epsilon) = u_U + \beta \Omega^U(\epsilon) \quad (16)$$

$$\Omega^U(\epsilon) = \mathbb{I}_{=0}^{RC,u} W_C(\epsilon) + \mathbb{I}_{=1}^{RC,u} \left[(1 - p(\theta))U(\epsilon) + p(\theta)\Omega^R(\epsilon) \right] \quad (17)$$

$$\Omega^R(\epsilon) = \mathbb{I}_{=1}^R W_R(\epsilon) + \mathbb{I}_{=0}^R U(\epsilon). \quad (18)$$

The non-rider searchers decide whether to take instantly a C platform job that yields a value $W_C(\epsilon)$ or search for a job with the R platform to achieve a value $W_R(\epsilon)$. In the latter case, due to frictions, they fail to receive a job with probability $1 - p(\theta)$, in which case they remain outside of the platform sector. Alternatively, with probability $p(\theta)$, they can choose between remaining outside of the platform sector or accepting the job offer in R . The policy indicator $\mathbb{I}^{RC,u}$ captures the non-rider's decision to search for R jobs rather than taking a C job, while $\mathbb{I}^R$ denotes their decision to accept an offer from the R platform. To reduce notation, we omit the dependence of policy functions on the taste parameter ϵ here and throughout the sequel.

Riders in the C platform may search for a job in the R platform. We introduce on-the-job search to capture the substantial overlap at relatively low hours in the distributions of hours worked for R and C jobs displayed in Figure 2. We interpret this fact as reflecting that riders with relatively high work dis-utility would still accept C jobs while keep on searching for better-paid, shorter-hours R jobs. Then, assuming that the job offer rate,

$p(\theta)$, is the same as for non-rider searchers, the resulting value becomes:

$$W_C(\epsilon) = u_C(\epsilon) + \beta \Lambda^C(\epsilon) \quad (19)$$

$$\Lambda^C(\epsilon) = (1 - p(\theta)) \Lambda^{CC}(\epsilon) \quad (20)$$

$$+ p(\theta) \left[\mathbb{I}_{=1}^{CR} \left(\mathbb{I}_{=1}^R W_R(\epsilon) + \mathbb{I}_{=0}^R U(\epsilon) \right) + \mathbb{I}_{=0}^{CR} \Lambda^{CC}(\epsilon) \right]$$

$$\Lambda^{CC}(\epsilon) = \mathbb{I}_{=1}^C W_C(\epsilon) + \mathbb{I}_{=0}^C U(\epsilon), \quad (21)$$

where $\mathbb{I}_{=1}^{CR}$ is an indicator of the rider preferring a R job over a C job offer, while $\mathbb{I}_{=1}^C$ denotes that the rider prefers a C job over non-employment. Likewise, $\Lambda^{CC}(\epsilon)$ captures the value of staying with the C job or moving to the non-rider option. Note that on-the-job search in R jobs is omitted as riders could always choose to get a job directly in the C platform. Hence, with $\Lambda^R(\epsilon)$ capturing the value of staying with the R platform or becoming a non-rider, the value for a rider in the R platform solves:

$$W_R(\epsilon) = u_R(\epsilon) + \beta \Lambda^R(\epsilon) \quad (22)$$

$$\Lambda^R(\epsilon) = (1 - \delta) W_R(\epsilon) + \delta U(\epsilon), \quad (23)$$

Having characterized labor supply choices from the value of each job, we now define the two types of individuals who search for jobs in the R platform. First, there is a mass of searchers who are currently non-riders, u , and are only willing to accept a direct job offer as a rider from the R platform. Second, there is a mass of on-the-job searchers who are currently working in the C platform, c , but would prefer transiting to the R platform. Thus, the total mass of job searchers, s , is a combination of the non-riders and C riders on-the-job seekers given by:

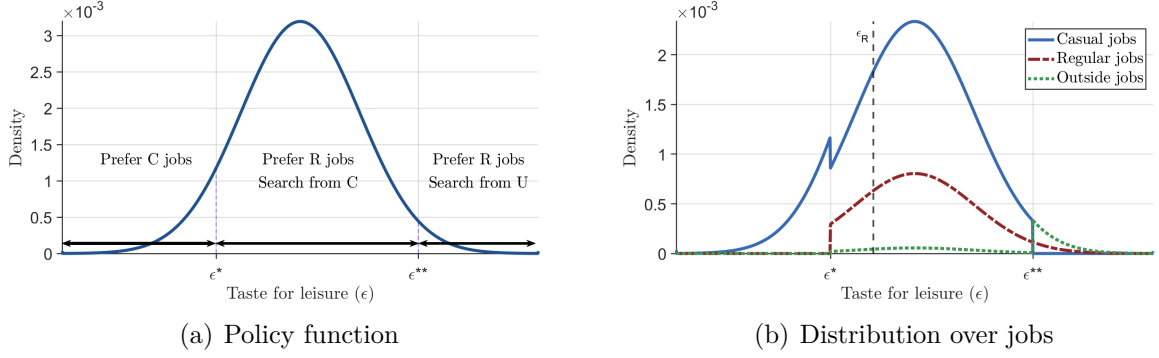
$$s = u \cdot \int_0^\infty \mathbb{I}_{=1}^R(\epsilon) \mathbb{I}_{=1}^{RC,u}(\epsilon) dG^{UR}(\epsilon) + c \cdot \int_0^\infty \mathbb{I}_{=1}^{RC}(\epsilon) dG^{CR}(\epsilon), \quad (24)$$

where, G^{UR} and G^{CR} are the CDFs of the taste parameter ϵ of the u and c searchers, respectively.

Equilibrium. The equilibrium consists of prices and quantities for food delivery services (p_R, p_C, o_R, o_C) , employment levels (n_R, n_C) , labor supply (h_C) , utilization rate (φ) , labor market tightness (θ) , mean leisure taste $(\bar{\epsilon}_R)$, wage in the R platform (w_R) , value functions $(J_R, J_R^v, U, \Omega^U, \Omega^R, W_C, \Lambda^C, \Lambda^{CC}, W_R, \Lambda^R)$, and distributions $(G^C, G^R, G^{UR}, G^{CR})$ such that:

- **Food delivery market.** Consumer demand (o_j, o_{-j}) maximizes utility as given by equation (5) for $j \in \{C, R\}$, while satisfying the budget constraint in (4). Moreover, platforms maximize profits and set prices according to (6).

Figure 3: Riders' preferences for jobs and employment distribution



Source: Own elaboration from simulated data.

Note: The left panel displays workers' policies as a function of their time preferences. The right panel displays the employment distributions of the two platforms over workers' time preferences.

- **Labor market.** Riders optimize labor supply h_C in C jobs as in (13). Workers' value functions satisfy (16)-(23); the value of a vacant job J_R^v satisfies the free-entry condition (9); the value of a filled vacancy J_R satisfies (8); the hourly wage w_R is set by Nash bargaining as in (12), and the utilization rate φ adjusts the demand for and supply of hours in C jobs following (2).
- **Distributions are consistent.** $(G^C, G^R, G^{UR}, G^{CR})$ arise from optimal policies and equilibrium outcomes, with G^R determining $\bar{\epsilon}_R$.

4.1 How the model works

Before examining the strategy to quantify the model parameters in detail, we first discuss riders' choices of their preferred job type and their resulting distribution across jobs. Panel (a) in Figure 3 shows the distribution of ϵ under the benchmark calibration for the model economy discussed in Section 5.1 below, along with the key cutoff values, ϵ^* and ϵ^{**} , determining employment choices. Riders prefer C jobs over other jobs whenever their idiosyncratic distaste for work is low enough, namely, $\epsilon < \epsilon^*$, where ϵ^* solves $W_C(\epsilon^*) = W_R(\epsilon^*)$. The insight is that C jobs help achieve high earnings by working many hours.

Conversely, riders prefer R jobs over other options whenever $\epsilon > \epsilon^*$, this time because they offer a wage premium relative to both C and outside jobs. Yet, due to search frictions, some of these riders (those without a too high taste for leisure) are willing to accept momentarily C jobs to then search on-the-job for their preferred R option. This happens whenever $\epsilon \in (\epsilon^*, \epsilon^{**})$, where the cutoff ϵ^{**} solves $W_C(\epsilon^{**}) = U(\epsilon^{**})$. Finally, workers with a very high taste for leisure, $\epsilon > \epsilon^{**}$, search for R jobs from outside jobs.

Panel (b) of Figure 3 displays the resulting employment distribution over C and R jobs in our calibrated model. It shows that most self-employed riders would prefer to

become employees. The figure also displays a dashed black line that represents the level of preferences for leisure, ϵ_R , above which riders currently operating for the C platform work fewer productive hours than in the R platform, $\varphi h_C(\epsilon_R) = \bar{h}$. In our calibration, this share is large: 72.7 percent of C riders work more hours than those at the R platform, but their productive hours are lower than in the latter. We note that this phenomenon arises because of the interaction of the search friction, which prevents an instant reallocation of those workers to the R platform, and the relatively low utilization rate at the C platform, which reduces their productive hours. Thus, a policy inducing worker reallocation from C jobs to R jobs, *ceteris paribus*, would be highly beneficial to these workers as they will enjoy higher earnings despite working fewer hours.

5 Model Calibration

Table 2 presents the choice of the model parameters. The key moments of the calibration strategy that allow us to evaluate the labor market and welfare consequences of the RL can be summarized as follows: First, we measure the size of the administrative sanctions brought about by the RL. Second, the calibration captures the riders' trade-off between higher net wages in R platforms and more flexible hours in C platforms. Third, given the relative desirability of the two types of platforms, the effect of the RL sanctions depends on (i) the relative size of the two platforms, (ii) how price changes resulting from sanctions lead to consumer demand shifts, i.e., the substitutability of platforms from a consumer point of view, (iii) how attractive it is for workers to take up jobs outside of the food delivery sector, i.e., the sector-level labor supply elasticity, and (iv) how riders' utilization rate at C platforms changes in response to demand changes. Our calibration targets the moments (i-iii), and Section 5.3 shows that the resulting model matches (iv). We begin by discussing those parameters directly from the data and then turn to those that are estimated by the Simulated Method of Moments (SMM).

5.1 Calibrated parameters

The calibration captures the post-reform labor market. The model period is a month, and the annual discount rate is set to 4.0 percent. We calibrate the administrative sanctions using annual reports from the two non-compliant platforms (Glovo and Uber Eats). To measure the present discounted expected value of sanctions, we use these firms' provisions for RL-related legal risks due to classifying riders as self-employed (see Appendix A). To obtain a moment comparable to our steady-state model, we compute the flow value of these provisions relative to net revenue. By 2024, expected sanctions reached 3.2 percent of net revenue, which requires $\Gamma = 0.04$.

Table 2: Summary of model calibration

Parameter	Description	Value	Moment
Panel I: Calibration			
<i>A: Sanctions</i>			
Γ	Administrative sanctions	0.04	Ratio sanctions to net revenue
<i>B: Preferences</i>			
$(1 - \beta) * 100$	1 – Discount factor	0.34	4% annual discount rate
μ	Elasticity of substitution	6.00	García-Perea et al. (2021)
<i>C: Labor market</i>			
$\bar{h}_R$	Hours worked in R	0.16	Mean hours in $R/24$
$\log(\kappa)$	Vacancy costs	7.03	Hagedorn and Manovskii (2008)
η	Workers' bargaining weight	0.50	Petrongolo and Pissarides (2001)
δ	Destruction rate in R	0.07	EU flow $R = 7\%$
<i>D: Outside option</i>			
$\bar{h}_U$	Hours outside option	0.16	Predicted hours worked/24
$\log(b_U)$	Flow value outside option	1.66	LS elasticity = 0.48 (Hall et al., 2023)
<i>E: Taxes</i>			
τ_c	Payroll taxes in C	0.16	Payroll taxes self-employed
τ_f	Payroll taxes in R	0.29	Payroll taxes employers
Panel II: SMM			
<i>A: Preferences</i>			
s_R/s_C	Utility shifter goods	1.62	Employment share of R platform
μ_ϵ	Work disutility: Mean	3.63	Mean hours in $C/24 = 0.22$
σ_ϵ	Work disutility: Std. dev.	0.66	95 th percentile hours in $C = 7/24$
<i>B: Technology</i>			
$\log(A)$	Hourly productivity	2.54	Mean log wages in $R = 2.07$
ι	Matching parameter	0.22	Wage premium in $R = 0.17$

Note: The Table describes the quantification of the parameters and their respective targets.

Regarding consumers' preferences, the elasticity of substitution between orders, μ , determines the size of markups. Since evidence for the food delivery sector in Spain finds that average markups by the different platforms range between 15 and 30 percent over the past years (Ortiz Peralta, 2025), we set $\mu = 6$.¹⁴

We set the number of hours worked in the R platform to the mean observed in the convenience survey. Regarding the hours in the outside option, we interpret them as representing alternative opportunities to those found in the food delivery sector, including non-employment. Accordingly, we first calibrate the number of hours worked in these jobs as the predicted hours of workers in the Spanish economy with the same observable characteristics as the average rider. In particular, we regress the annual hours worked

¹⁴The size of these markups is also consistent with aggregate evidence from the Spanish economy (García-Perea et al., 2021).

by an individual i , h_i , on a vector of characteristics, X_i , including sex, age, nationality, skills, occupation, and sector. To estimate this regression, we use annual microdata from the 2022 wave of the Wage Structure Survey (EES), which is the only wave available with information on hours worked for a representative sample of employees in Spain during our reference period. To determine the average sociodemographics of riders, $\bar{X}_i$, we draw data again from the convenience survey. The OLS-estimated parameters, $\hat{\beta}$, from that regression yields a prediction of $\hat{\beta}\bar{X}_i = 5.0$ hours worked for workers with the average characteristics of riders in similar occupations. We then adjust this number for the unemployment rate among such workers using quarterly data from the Spanish Labor Force Survey (LFS) for the last two decades. Our main finding is that the predicted unemployment probability for this specific type of worker is 24 percent. Assigning zero hours when unemployed, we obtain $\bar{h}_U = 0.76 \cdot 5.0 / 24 = 0.16$. Since we set the fraction of hours worked in the R platform to $\bar{h}_R = 0.15$, consistent with a mean daily hours of 3.7, our calibration strategy implies that riders expect to work similar hours both in outside jobs and the R platform.

Next, we turn to those parameters guiding labor market flows in the R platform. Following [Hagedorn and Manovskii \(2008\)](#), the flow vacancy cost κ is set to match an expenditure made by firms equivalent to forgoing 58 percent of average labor productivity per worker, and, in line with [Petrongolo and Pissarides \(2001\)](#), we set the bargaining power of riders to $\eta = 0.5$. Both values are in the ballpark of what is often used to calibrate frictional models of the Spanish labor market (see, e.g., [Bentolila et al., 2012](#)). Regarding employment separations, we choose the monthly job destruction rate, δ , to match a 7.0 percent transition rate from employment to non-employment in the Postal Services and Courier Activities sector (NACE 532H) of the largest Spanish administrative dataset (Muestra Continua de Vidas Laborales-MCVL) for the post-reform period September 2021-December 2022.¹⁵

As for payroll taxation, we choose tax rates in the C and R platforms to match the average payroll taxes of the self-employed, $\tau_c = 16$ percent, and the employers, $\tau_f = 29$ percent, respectively.

5.2 Jointly estimated parameters

The remaining six parameters $\mathbf{o} = (s_R, \mu_\epsilon, \sigma_\epsilon, \iota, A, b_U)$ are those related to consumers' preferences, riders' tastes for hours worked, the matching and production technologies, and the flow value of riders' outside option. To estimate these parameters, which jointly

¹⁵This NACE sector includes riders together with postal staff and other types of couriers. Since these workers have more stable labor contracts than riders, we restrict the sample to employees who work for two consecutive months and whose tenure is no longer than 18 months, to be consistent with the tenure reported in Table 1.

determine the distributions of market shares of two types of platforms, hours worked, wages, and employment rates, we use the SMM approach. Below, we describe each parameter and its closest associated moment. Moreover, Table C.1 in Appendix C reports the fit of the targeted moments, which turns out to be very satisfactory.

The utility shifter s_R is informative about the relative size of the R platform by affecting the relative demand for food delivery services, as shown in Equation (5). Therefore, following the results from the convenience survey in Table 1, 24 percent becomes the target for the employment share of riders in these platforms.

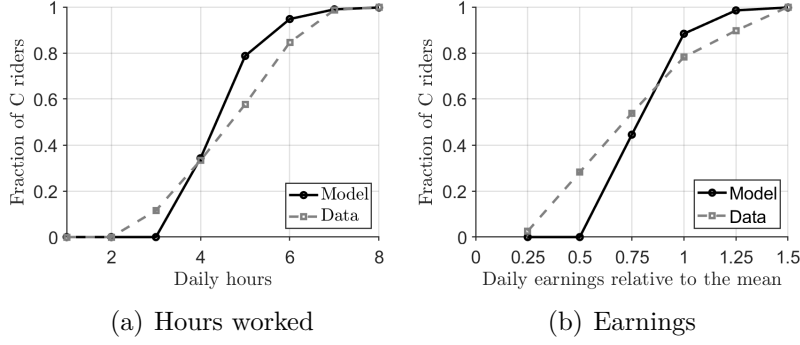
The distribution of work hours in the C platform is informative about riders' work tastes, ϵ , since self-employed riders choose their own working hours. Note that the model controls for the self-selection of workers with a strong taste for flexible hours into the C platform. In the data, we observe an average of 5.4 and a 95th percentile of 7.0 hours worked in the C platform. Thus, these moments are the ones used to estimate the mean, μ_ϵ , and standard deviation, σ_ϵ , of idiosyncratic preferences.

Next, consider the parameters related to production and matching technologies. Given the level of consumer spending, the productivity per hour, A , determines the marginal revenue productivity of riders and thereby affects hourly wages on both platforms. By influencing the matching probabilities, the parameter ι determines the size of the match surplus for R jobs, which influences the wage premium in the R platform relative to the C platform. To discipline these parameters, we target an average hourly wage net of social security contributions of €7.9 euros in the R platform, based on the sector's collective bargaining agreement.¹⁶ In addition, using once again the evidence from the survey, we also target a wage premium of 17.2 log points for R jobs.

Lastly, we turn to the expected flow value of riders outside the delivery sector (including non-employment), b_U . This parameter governs the labor supply elasticity to the platform sector by determining the cutoff taste for leisure of the marginal rider, ϵ^{**} (see Section 4.1). In particular, a worse outside option shifts the taste for leisure of the marginal rider to the right. Hence, fewer workers enter the platform sector in response to a marginal increase in its wages. Experimental evidence from exogenous changes in the compensation policies of ride-hailing firms points to a market-level elasticity of total hours supplied in the platform sector of about one-half (Hall et al., 2023; Caldwell and Oehlsen, 2026). Accordingly, we calibrate the outside option by matching the labor

¹⁶We use the [collective agreement](#) (M/441/2021/N), signed between Just Eat and trade unions in 2021 as a representative source of information for remuneration of R riders. We compute our estimate of hourly wages by dividing the annual base wage (€15,232), net of employees' social security and income taxes, by the annual number of hours for full-time riders (1,792). We find net hourly wages to be €7.8. This result is reassuring, as we find that net hourly wages of R riders in the convenience survey are around €7.0

Figure 4: Model fit of hours worked and earnings in C jobs



Note: The left panel of the Figure displays the cumulative distribution of hours worked among riders in C jobs in the data and model, and the right panel displays the cumulative distribution of daily earnings (relative to the mean) among C workers in the data and model. The model and data periods are post-Rider’s Law.

supply elasticity resulting from a labor demand shift induced by an expenditure shock ($\Delta \bar{m}$):

$$\varepsilon_H = \frac{\Delta \log H}{\Delta \log \bar{w}} = \frac{\log H^{\text{counterfactual}} - \log H^{\text{baseline}}}{\log \bar{w}^{\text{counterfactual}} - \log \bar{w}^{\text{baseline}}},$$

where H denotes total hours worked by riders, $\bar{w}$ the riders’ average hourly wage, and the counterfactual is the economy with higher labor demand. This calibration implies $\log(b_U) = 1.66$ to match an elasticity of $\varepsilon_H = 0.48$ (see Table 8 in Hall et al., 2023).

5.3 Model fit of untargeted moments

To validate the predictions of the model, we compare how the model fits untargeted moments of self-employed riders with regard to: (i) hours worked, (ii) earnings, and (iii) the pass-through of labor demand shocks to utilization rates.

A key feature of the data is that the hours distribution in C jobs is smooth and exhibits relatively high dispersion, which our calibration interprets as heterogeneous preferences for working hours among its riders. Reproducing these features of the data is policy-relevant because they reflect the value that workers place on their hours choices. Figure 4(a) displays the post-RL cumulative distributions of daily hours worked among C workers in the data and model, which turn out to be closely aligned. Specifically, the model reproduces that about one-third of C riders work at most 4 daily hours, while nearly one-tenth exceed 6 hours, and the majority work around 5 hours.

Another policy-relevant feature is the different importance of C jobs in terms of labor earnings for riders. Figure 4(b) displays their post-RL cumulative distribution among C riders both in the data and the model, where earnings are expressed relative to the mean. Two facts stand out. First, the model does a good job in replicating earnings

dispersion across riders. Note that, since it abstracts from hourly wage dispersion, this result indicates that the dispersion in hours worked alone is sufficient to rationalize earnings inequality among riders. Second, the model somewhat underestimates the share of riders with relatively low earnings; nevertheless, it correctly replicates that most riders earn within 25 percent of the mean.

Next, consider the effect of labor demand changes on C riders' utilization rates. Hall et al. (2023) provides direct empirical evidence on how self-employed platform workers' utilization rates respond to exogenous changes in compensation policies in the U.S. In particular, following a 1 percent Uber-initiated fare increase, the long-run change in hourly earnings is not statistically significant (coefficient 0.08, s.d. 0.06), whereas the utilization rate falls by about 0.70 percent, being significant (s.d. 0.07).¹⁷ Through the lens of our model, the log hourly pay of C workers is given by:

$$\log\left(\frac{y_C}{h_C}\right) = \underbrace{\log(\varphi)}_{\text{Utilization rate}} + \underbrace{\log((1 - \tau_c)A)}_{\text{Net earnings per ride}}$$

To replicate this quasi-experiment in our setup, we simulate a labor demand shock that raises earnings per ride one-to-one, comparing the benchmark with a counterfactual in which the C platform experiences a TFP shock, $\Delta \log(A_C) = 0.01$.¹⁸ The elasticity of the utilization rate is measured by comparing the counterfactual and benchmark economies:

$$\varepsilon_\varphi = \frac{\Delta \ln \varphi}{\Delta \ln(A)}.$$

The simulation yields an elasticity of $\varepsilon_\varphi = -1.0$, closely in line with the empirical evidence. In other words, an increase in C riders' earnings per trip does not lead to higher hourly pay, since the utilization rate declines almost one-to-one with the higher fare. Our model rationalizes this pattern: the utilization rate must fall for the market-clearing condition (2) to hold.

6 Policy experiments

This section quantifies the effects of the RL on the online food delivery sector by comparing our baseline (post-reform) economy ($\Gamma = 0.04$) with a counterfactual one where no fines are imposed on defier platforms ($\Gamma = 0$). Our model makes explicit how sanctions affect the structure of this market: relative to the counterfactual economy, sanctions in

¹⁷The estimates come from Tables 9 and 10 in Hall et al. (2023). Drivers earn a fraction of their fares (price per trip) and remit the remainder to Uber as the "Uber fee".

¹⁸The experiment from Hall et al. (2023) focuses on an unexpected change in fares from a *company-level* shock that hires self-employed workers. Thus, in this counterfactual, we keep the TFP level of the R platform constant ($\Delta A_R = 0$).

Table 3: The effects of the RL on the online food delivery sector

	No sanctions ($\Gamma=0$)	Baseline ($\Gamma=0.04$)		No sanctions ($\Gamma=0$)	Baseline ($\Gamma=0.04$)
<i>A: Orders</i>			<i>D: Employment</i>		
Relative price (p_C/p_R)	0.54	0.56	Employment C	0.77	0.73
Market share (o_C/o_R)	0.77	0.74	Employment R	0.22	0.24
Utilization rate (φ)	0.68	0.65	Outside jobs	0.01	0.03
<i>B: Wages</i>			<i>E: Hours</i>		
Mean log hourly wages C	1.98	1.93	Mean productive hours C	3.68	3.53
Mean log hourly wages R	2.13	2.10	Mean productive hours	3.69	3.57
Labor market tightness	0.29	0.32			
<i>C: Job preferences</i>			<i>F: Welfare change (%)</i>		
Cutoff C and R jobs, ϵ^*	2.82	2.70	Consumers	–	-3.01
Cutoff C and U jobs, ϵ^{**}	6.26	4.93	Riders	–	-4.17

Note: The Table displays the model results from the counterfactual simulation where defier, C , platforms do not have to pay a fine: $\Gamma = 0$.

the benchmark economy increase the marginal cost of using C riders, which in turn raise the relative prices of C orders, p_C/p_R , and shift consumer demand to R platforms, with a subsequent labor demand change across platforms.

6.1 Labor-market effects of the Rider’s Law

Before turning to the main counterfactual results, a few clarifications about the correct interpretation of these findings are in order. First, we abstract from the fiscal implications of the reform, since the resulting fines and changes in payroll-tax revenues are absorbed by the general treasury. Second, as stressed above, we also abstract from sector-wide demand trends, implying that our results should be interpreted as a counterfactual of how the riders’ sector would have fared absent the RL, holding those trends constant.

Panel A in Table 3 shows that administrative sanctions directly translate into a convergence of relative prices of orders, as C platforms raise prices to offset the increase in marginal costs and maintain profitability at the margin. We find that the relative price of C orders is 2.7 percent higher in the benchmark ($p_C/p_R = 0.56$) relative to the counterfactual without sanctions ($p_C/p_R = 0.54$). Though no detailed data on prices charged to customers for identical food deliveries is available, there is media evidence that Glovo (C) significantly charged higher commissions in the post-reform period, while those of Just Eat (R) remained at similar levels (see, e.g., [Qamarero, 2024](#)).

Note that a higher relative price of C orders shifts consumer demand to R platforms. In particular, we find that the RL increases the market share of the latter platform by 3.0 percentage points, which explains half of the post-reform change in this share observed in the data (see Figure 1).

The table shows that this shift in consumer demand decreases C rider's utilization rates, φ , from 0.68 to 0.65. This decline is not mechanic as the C sector provides fewer orders but riders also leave these platforms. To see which of the two forces dominate, suppose for the moment that the total number of riders is unchanged. In that case, as the rise in average order prices decreases total demand for orders, the utilization rate must fall when the riders switching from C platforms to R platforms produce more orders in their new jobs. The discussion in Section 4.1 highlights that this is indeed the case in our model. Hence, the only way for the utilization rate of C riders not to fall is that the total labor supply elasticity of the sector was high enough such that sufficiently many riders leave the sector, which is not the case in the quantitative model. As discussed in Sections 5.1 and 5.3, the quantitative model matches available evidence for platform workers on the labor supply elasticity and the response of utilization rates to income shocks which provides us confidence in our model prediction.

Panel B in Table 3 shows that, as the utilization rate among C riders falls, their mean net hourly wage decreases by 5.0 log points. This wage reduction induces a fraction of long-hours C riders to opt for R jobs, as illustrated by the reduction in the cutoff value ϵ^* in the baseline economy, from 2.82 to 2.70 (panel C). Overall, this leads to a fall of 4.0 percentage points in the employment share of the C platform (panel D). On average, sanctions reduce mean productive hours worked (panel E), due to riders' reallocation from the long-hours (C) to the short-hours (R) platform, as well as due to the falling utilization rate of stayers in the C platform.

The negative effects of sanctions on the C platform have meaningful spillover effects on the R platform. On the one hand, since working in a C platform becomes less attractive and the conditions of non-rider jobs are not affected by the RL, R riders' outside option worsens, weakening their bargaining position and pushing R wages down. On the other hand, the rise in the demand for R orders leads to a higher price, p_R , which is partially passed through to higher wages as their marginal revenue product of labor increases. Taken together, we find that the worsening of the outside option is the dominant channel, leading to a decrease of 3.3 log points in the R wage compared to the no-sanction scenario (panel B). As a result, sanctions do not change the R platform's wage premium much. Although microdata on wages for the pre-reform period is not available, there is some aggregate information about net hourly wages from the main C and R platforms. In 2019, Glovo paid their self-employed riders between €3.5 to €4 while Just Eat paid €5,

once social security contributions and other expenses are deducted. Thus, these data suggest that the wage premium in favor of R riders was also around 15 percent before the RL (see, e.g., [El País, 2020](#)).

Higher demand for R orders boosts vacancy creation and raises labor market tightness, θ , increasing the R employment share by 2.0 percentage points, from 0.22 to 0.24 (panel D). Overall, the R platform absorbs nearly half of the employment decline in the C platform due to sanctions, with the other half moving to the outside jobs (panel D). Thus, the overall sector shrinks by almost 2 percent. Put differently, the effects of the RL on total employment are negative but not large.

6.2 Welfare effects of the Riders' Law

We measure welfare in terms of consumption-equivalent variation, that is, the percentage of lifetime consumption that an individual would be willing to forgo to remain indifferent between the benchmark economy with sanctions and the counterfactual without them. The RL affects this welfare through changes in household consumption and riders' labor income and hours worked.

As reported above, the RL slightly contracts the food delivery sector, thereby reducing consumer demand in comparison to the counterfactual scenario. Moreover, as order prices from C platforms rise more than prices from the R platform, there is a reallocation of consumption across imperfectly substitutable orders. Consequently, we find that sanctions reduce consumer welfare from food delivery by about 3.0 percent.

On the flip side, rising prices have the potential to increase riders' welfare, as the sector as a whole produces fewer orders and, hence, requires fewer productive hours for the same total expenditures, m . In addition, as shown in Section 4.1, many workers with C jobs would prefer to transit to the R platform after the RL. In fact, in partial equilibrium, the reallocation towards employees would be beneficial, as it allows these riders to work fewer hours at R platforms while enjoying higher pay. However, in general equilibrium, the RL leads to an average-rider welfare loss of 4.2 percent.

These welfare losses depend on riders' idiosyncratic preferences over working hours, as shown in the left panel of Figure 5. Specifically, we define three groups of riders based on the cutoff values $\epsilon \in \{\epsilon_1, \epsilon_2\}$, such that,

$$W_R^{\text{RL}}(\epsilon_1) = W_C^{\text{RL}}(\epsilon_1), \quad \text{and} \quad W_C^{\text{RL}}(\epsilon_2) = (1 - p(\theta))U^{\text{RL}}(\epsilon_2) + p(\theta)\Omega_R^{\text{RL}}(\epsilon_2),$$

where the superscript "RL" in the riders' value functions indicates that the economy operates with sanctions for violating the RL. First, the Figure shows that nearly one-tenth of riders always prefer the C platform due to their low disutility from hours worked

($\epsilon < \epsilon_1$). These riders are the ones suffering the largest welfare losses, as they bear a substantial pay reduction in the C platform with no offsetting gains from improved job opportunities in the R platform.

Second, about 86.8 percent of riders, with $\epsilon \in (\epsilon_1, \epsilon_2)$, search for R jobs from C jobs under the RL. Sanctions generate two opposing effects for this group. On the one hand, these riders benefit from the sanctions because they facilitate transiting from C jobs to R jobs. On the other hand, they experience earnings and consumption losses because: (i) C platform's wages decline, hurting them while searching for an R job, and (ii) R platform wages also decline due to a lower option value in the C platform. The second channel dominates, so these riders are worse off on net.

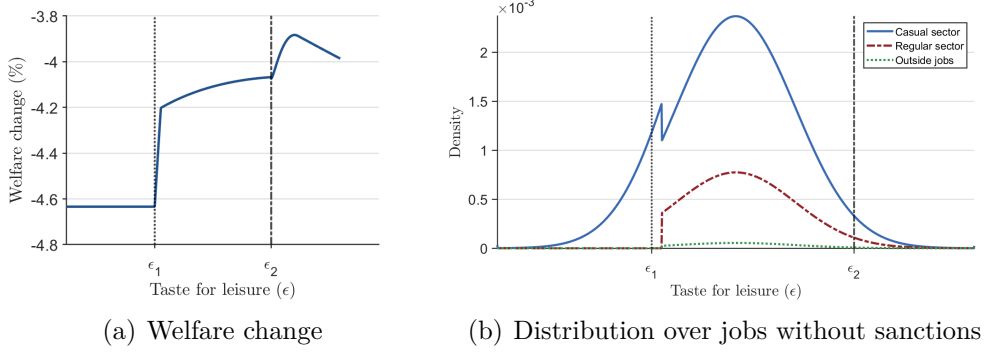
Third, the remaining 2.3 percent of riders, those with $\epsilon > \epsilon_2$, search for R jobs from outside jobs under the RL. As with the second group, the RL lowers the wages these riders receive upon finding an R job. Yet, they do not suffer from the worsening conditions of C jobs induced by the RL. Hence, their welfare losses are the smallest.

A relevant question is whether the same reform would have increased welfare if the riders' market had been structured differently. We find that the answer is affirmative when the riders' utilization rate in the C platform does not decrease excessively, so that their hourly wage does not fall by much. A substantially higher flow value of outside jobs ($\ln(b_U) = 1.79$) would achieve exactly this.¹⁹ We provide details on this thought experiment in Appendix D and only briefly sketch the intuition here. A better outside option implies that fewer riders are employed in the C platform with productive hours below those in the R platform prior to the reform ($\varphi h_C < \bar{h}_R$), as riders with a high preference for leisure now prefer more attractive outside jobs than working in the C platform. As discussed earlier, having only few unproductive workers in the C platform prior to the reform contributes to a small decrease in the utilization rate when sanctions are imposed. However, this alternative view of the food delivery market is inconsistent with the data in two key dimensions: First, displacing riders willing to work short hours from C jobs into outside jobs leaves no overlap in hours worked across platforms. Second, the sector-level labor supply elasticity becomes too elastic.

Lastly, recall from Table 1 that about 18.0 percent of riders lack work permits. One may argue that politicians are less concerned about their employment prospects because these workers cannot vote in elections. As these riders work particularly long hours, i.e., they are those endowed with low ϵ , their weight in our welfare calculations is relatively high. However, even ignoring the 18 percent of workers with the lowest ϵ from our welfare calculations, the RL still leads to a 3.5 percent welfare loss among riders.

¹⁹In this thought experiment, we simultaneously increase the value of R jobs with the utility shifter (s_R/s_C) for this platform to exist in equilibrium with a better outside option.

Figure 5: Welfare effects of the Riders' Law



Note: The left panel of the Figure displays the welfare effect of the Riders' Law (relative to the counterfactual economy without sanctions) in the vertical axis against workers' taste for leisure in the horizontal axis. The right panel shows the distribution of riders over the different types of jobs in the counterfactual scenario without sanctions.

6.3 Complementing the Riders Law with transfers and subsidies

So far, it has been implicitly assumed that any fiscal effects of the RL, including the sanctions, are absorbed by the general treasury. While this assumption is consistent with observed practice, one may ask whether the reform would be less detrimental if the revenue raised from the sanctions were redistributed to riders. We simulate a lump-sum transfer to all riders as well as a subsidy only to riders on the R platform through a reduction in payroll taxes.

6.3.1 Lump-sum transfers

We implement a lump-sum transfer T that satisfies the budget balance constraint, $T = o_C \cdot \Gamma$. This transfer reverts the welfare effects of the RL: under budget-neutral lump-sum transfers, welfare for the average rider increases by 0.47 percent instead of falling by 4.2 percent. Put differently, the average welfare loss among riders stems from the absence of riders' compensation during the reform rather than induced inefficiencies. Nevertheless, two additional issues are worth noting. First, since lump-sum transfers do not alter decisions by firms and workers, the food delivery sector still shrinks and consumer welfare still declines. Second, even a transfer that preserves average welfare does not avoid that riders staying in the defying C platform are worse off due to lower wages (see Figure C.1 in Appendix C).

6.3.2 Reducing payroll taxes

Instead of a lump-sum transfer, perhaps a more natural policy for the government may be to complement sanctions with lower social security contributions for R riders, as such a policy is more directly targeted at the jobs it intended to strengthen. To understand

the potential scope of this policy, we study a full-exemption of employer-paid taxes in R , decreasing the tax rate from 29 to 0 percent. This tax cut decreases marginal costs of the R platform and, thus, leads to a 6.4 percent increase in the relative price of C orders (p_C/p_R). To meet the additional demand, R firms increase vacancy creation, driving labor market tightness and wages up by 43.1 percent and 1.4 log points, respectively.

Nevertheless, relative to the scenario without sanctions, riders' average welfare still drops by 2.8 percent, and consumer welfare falls by 1.5 percent. Thus, complementing the RL with payroll tax cuts for R employers would have attenuated the adverse outcomes of the reform but not reverse them. The main insight for this result is that the R sector does not expand enough: the wage gain in R fails to offset the wage losses among C riders, who constitute the majority of riders, and the food delivery sector as a whole still shrinks due to the sanctions.

7 Conclusions

This paper quantifies the impact on employment, hours worked, and wages of policies mandating the use of dependent employees by online food delivery platforms. We focus on Spain, which is a forerunner in implementing such policies through the so-called Riders' Law (RL), enacted in 2021. This reform established the presumption of dependent employment for workers in this sector, known as riders, and levies large fines on platforms not complying with this legislation. Using information on defier platforms' provisions for RL-related legal risks, we estimate that sanctions account for 3.2 percent of their net revenue. A counterfactual simulation shows that the employee share would have been two percentage points lower in the absence of sanctions. Although the RL significantly incentivizes the employee status for riders, sanctions lead to an average welfare loss among riders of 4.1 percent due to lower wages, especially among those who remain working as bogus self-employed for the defier platforms after the reform. Most of this welfare loss stems from the fall in the utilization rate of C riders resulting from the reform. However, a lump-sum rebate to all riders of the amount of revenue raised by sanctions reverses their welfare loss and achieves a welfare gain of almost 0.5 percent for the average rider. At any rate, both with and without transfers, consumers suffer a welfare loss due to higher prices and lower orders, as well as a welfare loss of riders who remain as self-employed.

To reach these conclusions, we develop a quantitative equilibrium model with heterogeneous workers who sort into self-employed and employee jobs, each offered by a different platform that provides services to a representative consumer. These jobs are heterogeneous in terms of employability, work schedules, utilization rates of working time, and wages, capturing the main trade offs in the policy debate. The model is calibrated using several data sources, including a convenience survey, platforms' annual reports, and

aggregate statistics from the food delivery sector. This calibration implies that riders with relatively low tastes for leisure choose self-employed jobs because they offer greater flexibility in working long hours and are instantly available. By contrast, the remaining riders prefer to work for platforms offering employee contracts due to its wage premium, though they may accept to be self-employed or work in jobs outside the platform sector while searching for their preferred jobs.

Our finding that fines imposed on defier platforms are partially passed through to workers via lower employment, wages, and ultimately welfare should not be surprising. Yet, our model allows for two equilibrium mechanisms softening the blow to riders' labor outcomes, which, in principle, could be welfare-enhancing. First, resulting from higher wages and fewer hours worked as employees, shifting employment to these jobs improves, *ceteris paribus*, riders' welfare. Second, as fines affect the marginal costs of platforms, firms shift part of the burden to consumers/restaurants through higher prices, which could also benefit riders. We show that two key reason for the decrease in welfare are: (i) the fall in the utilization rate of self-employed riders who remain at the defier platforms jobs, and (ii) the reduction in the bargained wage at the complier platforms as riders' outside option deteriorates.

Finally, as we highlight differences in wages, hours, and frictions between the two types of platform jobs, we see several promising avenues for future research that extend the model to include other regulatory differences between these jobs, such as overtime pay, maximum hours regulations, vacation days, or health protection benefits.

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Appendix

A Estimating the size of government sanctions

This section documents the data sources and calculations underlying the estimation of the administrative sanctions from the RL. We use data from Glovo and Uber Eats, the two defier platforms that remained in the Spanish food delivery market after the RL.

Revenue of the *C* platforms in Spain (2021–2023). Table A.1 summarizes the resulting estimates of Glovo’s revenue in Spain. The 2021 figure (€203.8 million) corresponds to the consolidated revenue of Glovoapp Spain Platform S.L. as filed with the Commercial Registry (*Registro Mercantil*). For 2022 and 2023, following Glovo’s acquisition by Delivery Hero, we estimate Glovo Spain’s revenue as a share of Delivery Hero’s published consolidated net revenues for the European region (€980.5 million in 2022 and €1.522.4 million in 2023), consistent with Glovo Spain’s weight of 36 percent in terms of net revenue in 2021 (Delivery Hero SE, 2022, 2023, 2024). We estimate a lower and upper bound based on a deviation of 10 percentage points with respect to this baseline number.

Administrative sanctions (2021–2024). Our analysis focus on the amount of administrative sanctions that influence the decision-making of defiant firms through affecting their operating costs. Two facts are especially relevant for our purposes. First, these sanctions increase platforms’ variable costs. Specifically, the Labour Inspectorate and Social Security (*Inspección de Trabajo y Seguridad Social*, ITSS hereafter) sanctioned Glovo for subcontracting 37,348 riders as independent contractors, reaching €205.3 million in total as a result of the sum of a €123.5 million from a penalty for the labour-law violation (*acta de infracción*) and €80.0 million sanction from back-payment of avoided Social Security contributions (*acta de liquidación*) (Público, 2023). Since Uber Eats subcontracted fewer riders, this company accumulated only €23.2 million in sanctions from the ITSS during the same period, comprising a penalty for labour-law violations (€19.75 million) and back-payment of Social Security contributions (€3.45 million).

Second, the sanctions by the ITSS do not capture the present discounted expected value of sanctions faced by both platforms, which is the object of interest in our analysis. The reasons are that: (i) platforms could accumulate even more sanctions in the future, and (ii) the Spain’s National Court (*Audiencia Nacional*) did not yet confirmed the sanctions during the reference period (Casado, 2025).²⁰

²⁰For instance, the first sanctions against Uber Eats were not upheld by the courts until March 2026 (Borja, 2026).

Table A.1: Provisions for RL-related legal risks and net revenue (2024, million of euros)

	Glovo	Uber Eats
Provisions	492.2	171.3
Net revenue	681.1	136.1

Note: For Uber Eats, we use consolidated net revenue (*importe neto cifra de negocios*) reported by Portier Eats Spain S.L. in its annual accounts, which is the legal entity through which Uber Eats operates in Spain (Portier Eats Spain, 2025). For Glovo, we combine figures from annual reports with estimates. For 2021, we use consolidated net revenue reported by Glovoapp Spain Platform S.L. After this year, Glovo merged into the Delivery Hero group. Thus, for 2024, we estimate Glovo Spain’s revenue as a share of Delivery Hero Group’s European segment revenue, as reported in its Annual Reports (Delivery Hero SE, 2025). This share (36%) is derived from the ratio of Glovoapp SA’s revenue to Delivery Hero’s European segment revenue in 2021 (Delivery Hero SE, 2022), the last year prior to Delivery Hero’s acquisition of Glovo; specifically, $203.8/571.4 \approx 0.36$. Regarding the source of provisions, Delivery Hero SE (2025) establishes “Delivery Hero has provisioned €492.2 million for legal risk related to rider reclassification and connected VAT risks”. Portier Eats Spain (2025) increased its provisions from 0 by December 2020 to €136.3 million by December 2021, which further increased to €171.3 million by December 2024.

To measure the present discounted expected value of sanctions, we turn to the non-compliant firms’ provisions for RL-related legal risks for classifying riders as self-employed. Table A.1 reports the amount of these provisions by December 2024 according to their annual reports. In total, defiant platforms accumulated €923.5 million in RL-related provisions for legal risks up to December 31, 2024.

Estimate of sanction-to-revenues. Since our model is in steady-state, we compute the flow value of these sanctions using a discount factor of 0.96, implying a total flow sanction of €26.5 million among non-compliant platforms. To obtain a comparable moment within the model, we express the flow value of these provisions relative to non-compliant platforms’ net revenue as of 2024, which yields:

$$\frac{\text{Flow value of sanctions}}{\text{Net revenue}} = \frac{26.5}{817.2} = 0.032.$$

Thus, we estimate that expected sanctions reach 3.2 percent of net revenue for C platforms. This is the target in our calibration exercise, leading to $\Gamma = 0.04$ in the model. Note that this likely represents a lower bound for the underlying sanctions of the reform, as Glovo Spain likely represents a smaller fraction of Delivery Hero’s net revenue than the one we use in our estimates (e.g., this group has acquired more firms over time).

B Convenience survey

We collected data on the labor market conditions in the food delivery sector through a convenience survey of food-delivery riders fielded in September 2023. We provide below the structure of the convenience survey for the blocks of wages and hours worked. The

questionnaire was administered in Spanish; the text reproduced is our English translation.

Questionnaire

RIDERS' PERSONAL AND JOB CHARACTERISTICS

Since the Covid-19 pandemic, the fast-food delivery rider sector has become an increasingly popular option for some workers due to technological advancements and flexible schedules. However, given the lack of information on these work practices and the potential consequences of the Rider Law 12/2021, the objective of this study is to determine its impact through this survey.

YOUR COOPERATION IN COMPLETING THIS SURVEY IS GREATLY APPRECIATED. Please note that the data you provide in this survey is protected by the European Union's General Data Protection Regulation (GDPR), which guarantees the confidentiality and privacy of the information collected. The survey is completely anonymous and complies with GDPR regulations on the protection of personal data.

QUESTIONNAIRE. Please fill in the requested information in the rectangle below each question.

Block H: Wages

H.1. Report your total gross pay (money going to your pocket before paying social security contributions or maintenance costs—see below) in Euros (for example, 1000, 1350, etc.) received from your platform during the last completed month prior to taking this survey. Exclude tips and ignore income taxes (IRPF) in reporting gross wages.

H.2. If you work as a rider for the Glovo or Uber Eats platforms, report your monthly vehicle and maintenance total expenses in Euros during the last completed month prior to taking this survey. These items include: (i) purchase, fuel, and repairs of your bicycle or motorcycle; (ii) bills on the cell phone and internet data usage required for continuous use of the application; (iii) cost of the thermal backpack and appropriate clothing for rain or cold weather; and (iv) payment of the self-employed monthly social security contribution.

Block I: Hours of Work

I.1. Report your daily hours of work (both riding on the road and waiting for orders) at your platform on a normal day of the week prior to taking this survey (for example, 2h, 4h, 9h, etc.).

I.2. Report how many days did you work in the week prior to taking this survey.

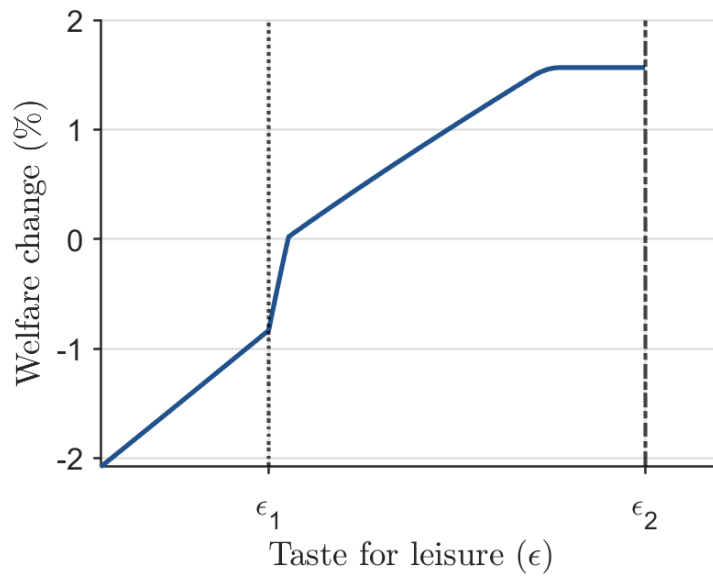
C Additional figures and tables

Table C.1: Model fit of estimated parameters

Parameter	Moment	Model	Data
<i>A: Preferences</i>			
s_R/s_C	Employment share of R platform	0.24	0.24
μ_ϵ	Mean hours in $C/24 = 0.22$	5.44	5.40
σ_ϵ	95 th percentile hours in $C = 7/24$	7.03	7.00
<i>B: Technology</i>			
A	Mean log wages in R	2.10	2.07
ι	Wage premium in R	0.17	0.17
<i>C: Outside option</i>			
b_U	Labor supply elasticity	0.48	0.48

Note: The Table reports the values of the SMM approach targeted moments in the model and the data.

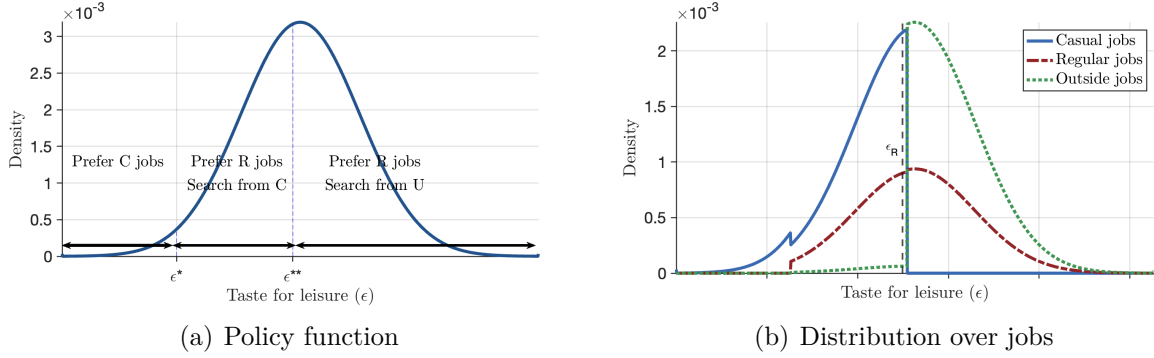
Figure C.1: Welfare effects of the RL with welfare-neutral lump-sum transfers



Source: Own elaboration from simulated data.

Note: The Figure displays the welfare effect of the RL complemented with lump-sum transfers ($T = \Gamma o_C$, and $\Gamma = 0.04$) as a function of workers' time preferences (ϵ).

Figure D.1: Riders' preferences and employment distribution in the alternative economy



Source: Own elaboration from simulated data.

Note: The Figure displays the employment distributions of the two sectors over workers' time preferences. Note that this is the economy where the parameterization remains the same as in the benchmark except for $\ln(b_U) = 1.79$ and $s_R/s_C = 3.0$.

D Understanding the welfare effects of the RL

This section discusses how our model allows for the possibility that the RL could be welfare-improving. To do so, we parameterize an alternative model economy in which the RL is beneficial and then show that the data lead us to reject this parameterization.

We focus on the minimum parameter adjustments required for the RL to yield a welfare gain slightly above zero for the average rider. A key condition for this to happen is that the wage in the C platform does not deteriorate much, namely, the utilization rate does not decrease excessively. For that to be the case, the technological constraint in Equation (2) establishes that

$$d \log(\varphi) \approx 0 \iff d \log(o_C) \approx d \log(n_C \cdot \mathbb{E}(h_C(\epsilon))).$$

In other words, the percent decline in orders must be similar to the percent decline in total hours worked following the RL. To achieve this outcome, our alternative market structure keeps the same fundamentals as the benchmark calibrated economy except for two parameters. First, we increase the flow value of the outside option to $\ln(b_U) = 1.79$. A high outside option implies that nearly all riders in the C platform have productive (φh_C) and actual (h_C) hours above those in the R platform before the reform, as riders with a high preference for leisure choose to search for R jobs from outside jobs rather than from the C platform. As discussed in Section 6.1, this contributes to a small decrease in the utilization rate of their work time when sanctions are in place. Second, to ensure that riders still choose the R platform under a higher outside option, the value of these jobs must increase. This is achieved by increasing the utility shifter of R goods, $s_R/s_C = 3.0$.

Before discussing these counterfactual results, we first show that the data reject this

Table D.1: The effects of the RL in the alternative economy

	No sanctions ($\Gamma=0$)	Baseline ($\Gamma=0.04$)		No sanctions ($\Gamma=0$)	Baseline ($\Gamma=0.04$)
<i>A: Orders</i>			<i>D: Employment</i>		
Relative price (p_C/p_R)	0.38	0.39	Employment C	0.34	0.32
Market share (o_C/o_R)	0.58	0.55	Employment R	0.28	0.29
Utilization rate (φ)	0.70	0.70	Outside jobs	0.37	0.40
<i>B: Wages</i>			<i>E: Hours</i>		
Mean log hourly wages C	2.01	2.01	Mean effective hours C	4.23	4.25
Mean log hourly wages R	2.25	2.25	Mean effective hours	4.00	4.00
Labor market tightness	0.43	0.44			
<i>C: Job preferences</i>			<i>F: Welfare change (%)</i>		
Cutoff C and R jobs, ϵ^*	2.31	2.26	Consumers	–	-2.7
Cutoff C and U jobs, ϵ^{**}	3.61	3.55	Riders	–	0.1

Note: The Table displays the model results from the counterfactual simulation where defier firms in the C sector do not have to pay a fine: $\Gamma = 0$. This is the economy where the parameterization remains the same as in the benchmark except for $\ln(b_U) = 1.79$ and $s_R/s_C = 3.0$.

alternative view of the market in key dimensions. Figure D.1 displays the distribution of riders' preferences for jobs and the resulting employment distribution in this alternative economy. As noted above, increasing the flow value of the outside option displaces riders willing to work short hours from the C platform to outside jobs. This implies that (i) C riders work longer hours than their R counterparts, that is, they have $\epsilon < \epsilon_R$, and (ii) the marginal rider has a smaller taste for leisure. The resulting equilibrium therefore features no overlap in hours worked across platforms. The economy also displays an unrealistically high labor supply elasticity: Table D.1 below shows that the RL induces a gross employment change of 5 percent while wages remain unchanged. Neither feature holds in the data for the food delivery sector.

Next, Table D.1 reports the labor market effects of the RL in this alternative model economy. The RL induces the reallocation of consumption across platforms by increasing the relative prices of orders served by the C platform. Since the relative decline in hours worked and orders almost match each other, the utilization rate hardly changes. In particular, the utilization rate (φ) and wages in the C platform only drop by -0.005 percent. Moreover, the wage of the R platform also hardly changes because this platform expands, and the outside opportunities of R riders do not deteriorate enough. Overall, the market structure implies that the wage gains in the R sector compensate for the losses in the C sector, so average welfare increases by 0.1 percent. In other words, the RL is solving a clear inefficiency: riders could work similar hours at higher wages. However, for

this to hold, the market structure observed in the data should be completely different to what this simulation exercise predicts.